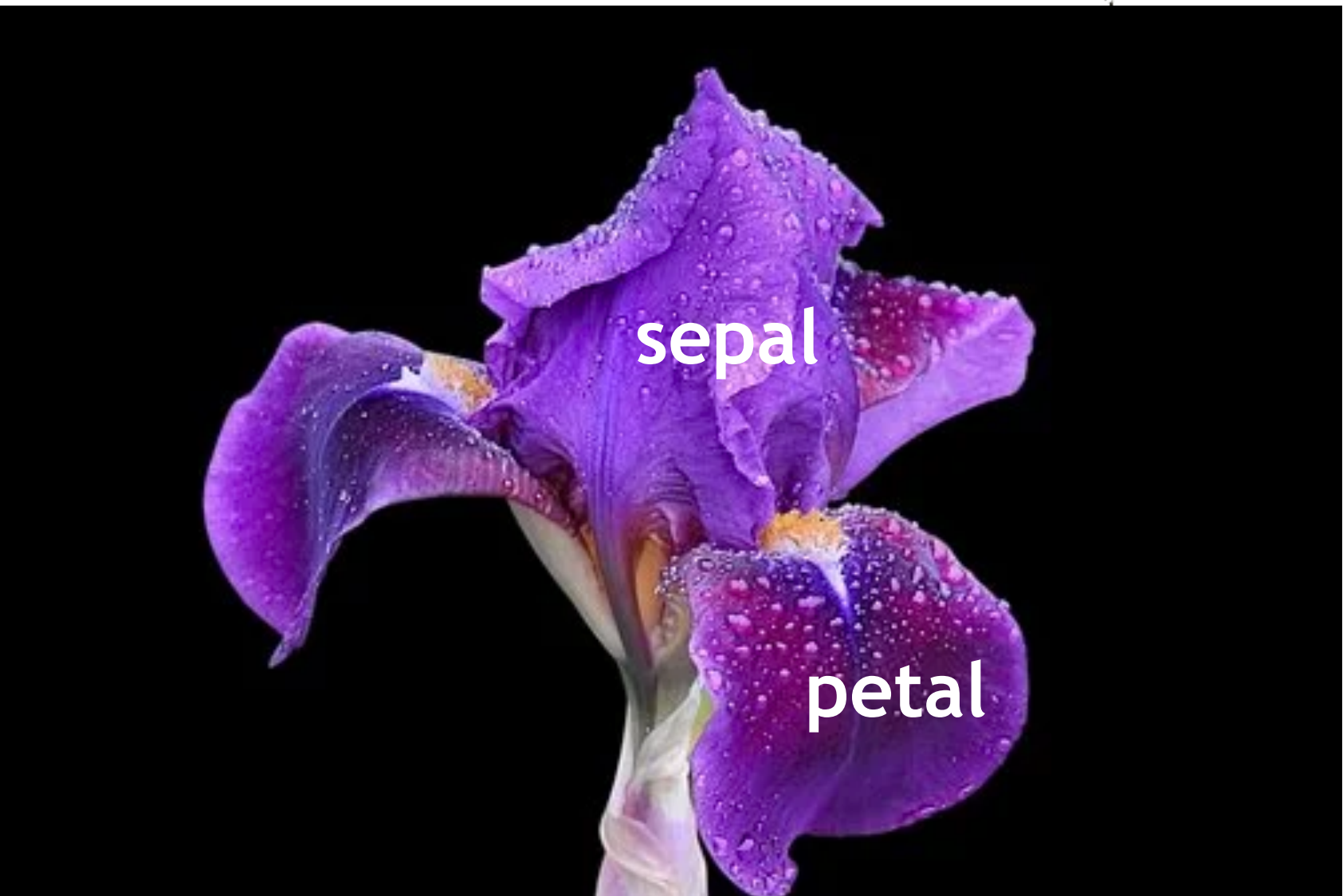


Continuous features



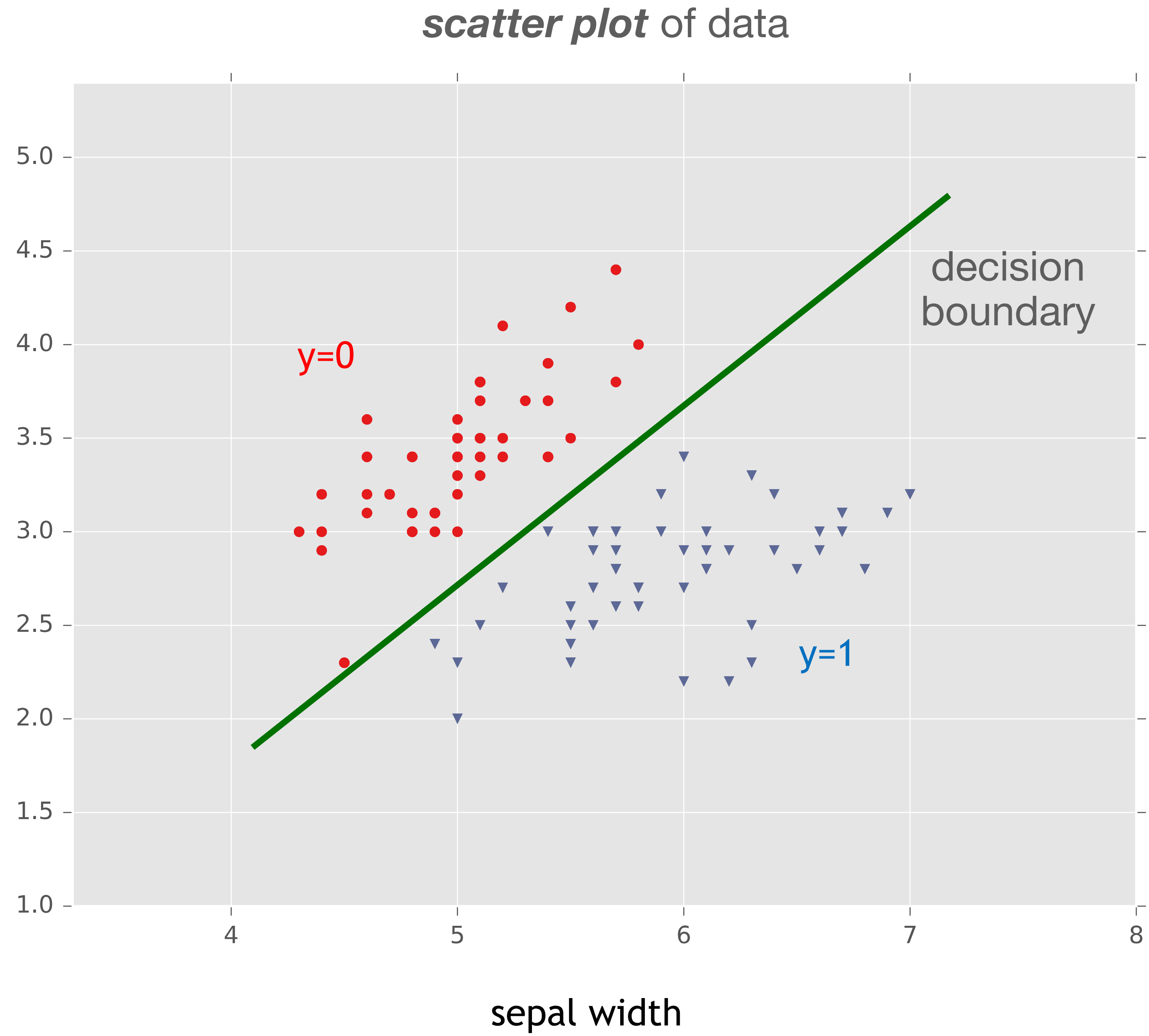
sepal length



Continuous features



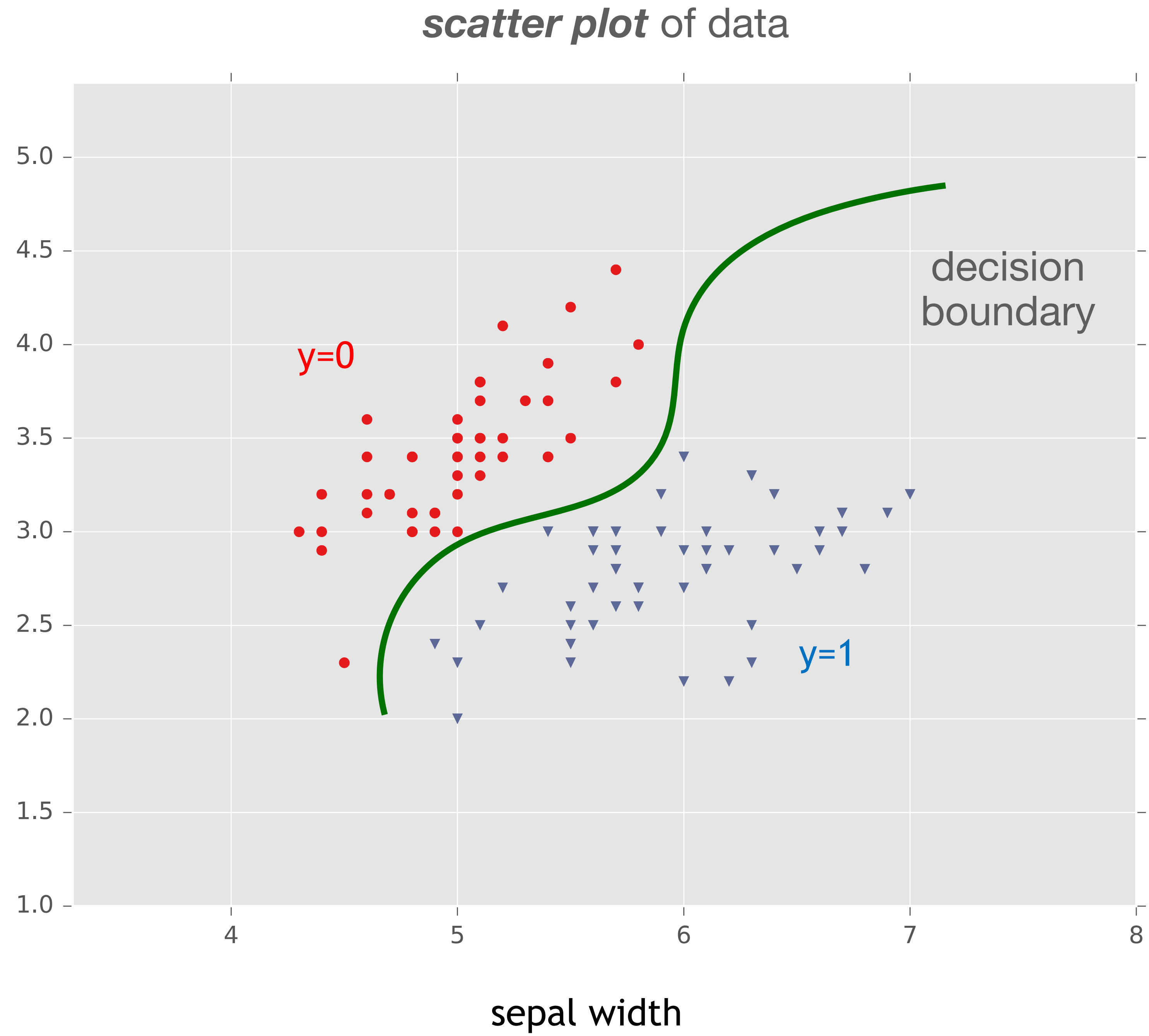
sepal length



Continuous features



sepal length



Overfitting



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Overfitting

- For *majority*, train error $\approx$ test error, while for *memorizer*, train error $\ll$ test error
 - ▶ we say that *memorizer* suffers from **overfitting**
- Overfitting = test error – train error
 - ▶ more generally, $P - \hat{P}$ (for min) or $\hat{P} - P$ (for max)
 - ▶ lots of overfitting means the difference is large
 - ▶ nothing to do with P or $\hat{P}$ being large or small on its own
- Heuristic: “simpler” learning methods tend to overfit less
 - ▶ we’ll try later to measure simplicity — not simple to do!
 - ▶ e.g.: complexity = # nonzero weights in a linear classifier, or complexity = how long we train a deep net

Underfitting

- On the other hand, *majority* doesn't seem like a great learner — not very expressive, high error
- This is called ***underfitting***
 - ▶ overfitting is absent or minimal, and another method could achieve better $\hat{P}$ while still limiting overfitting
 - ▶ heuristically, we kept the hypothesis class *too* simple

Why does overfitting happen?



Expected value of each die = 3.5

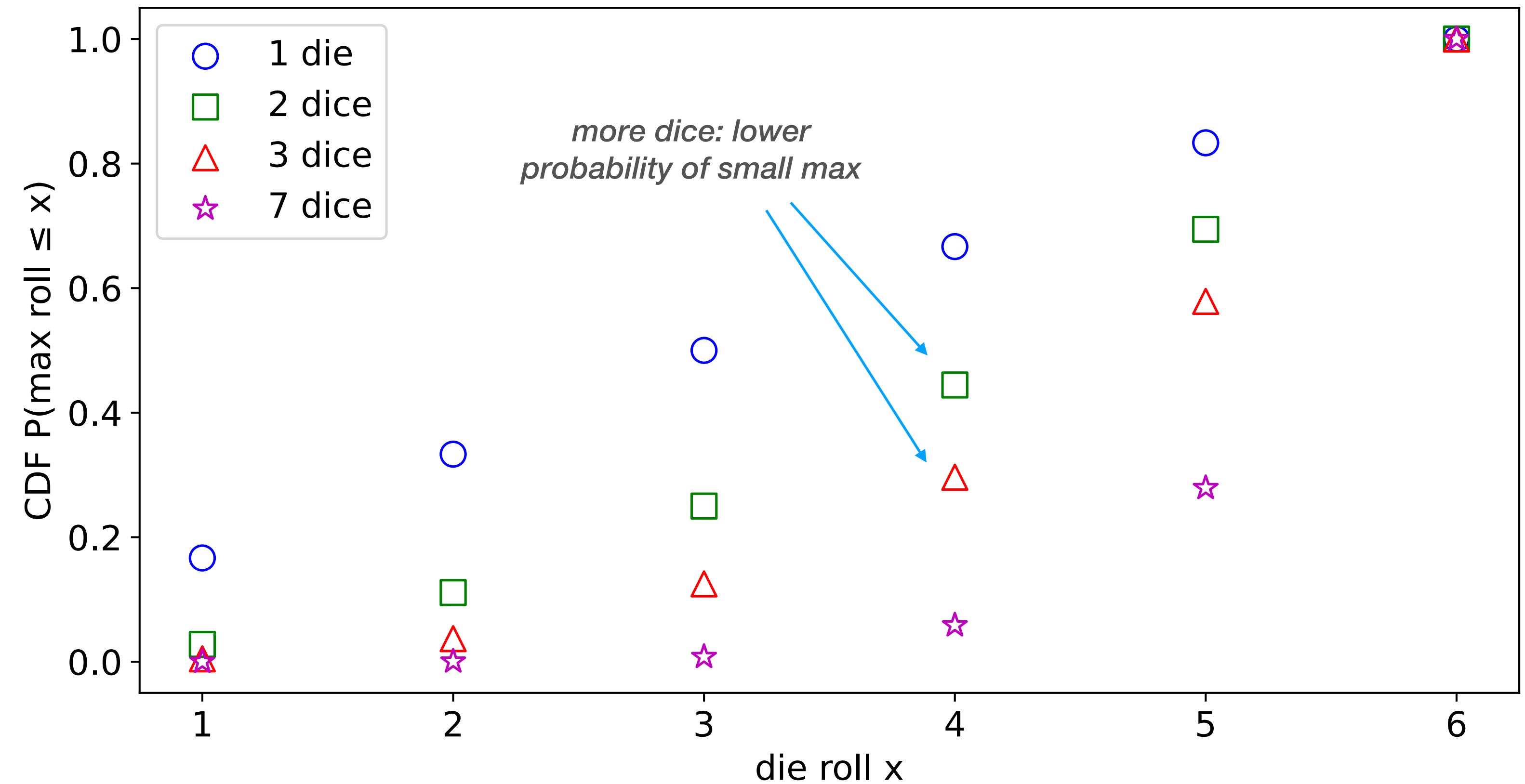
Why does overfitting happen?



Expected value of highest die =
lower than 3.5?
the same: still 3.5?
higher than 3.5?

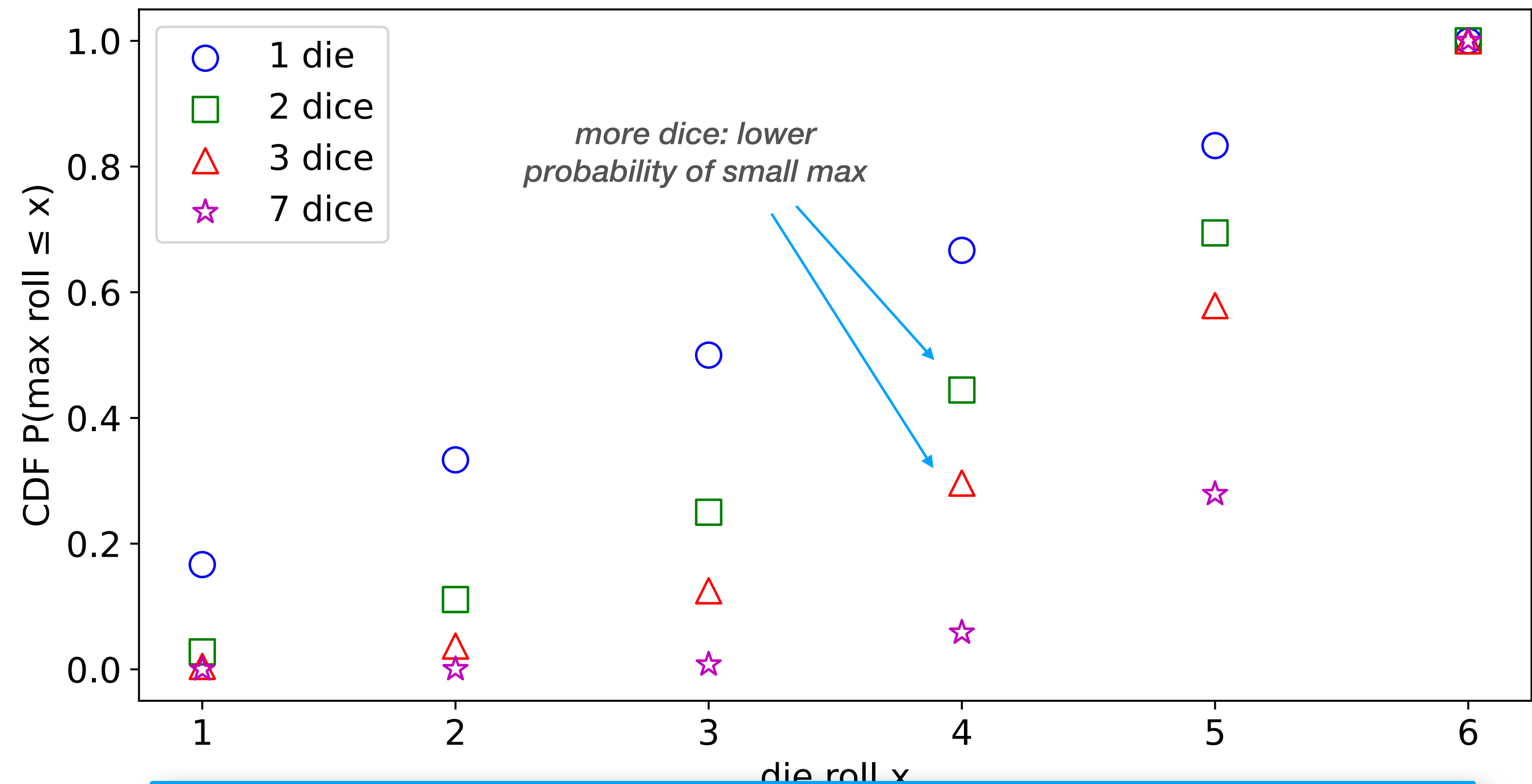
Expected value of each die = 3.5

Max of k dice



- CDF of one die: $i/6$
 - ▶ e.g., probability it's ≤ 3 is $\frac{1}{6} + \frac{1}{6} + \frac{1}{6}$
- CDF of max of k dice: $(i/6)^k$
 - ▶ probability of (all $3 \leq i$) is $P(\text{die } 1 \leq i) \cdot P(\text{die } 2 \leq i) \cdot P(\text{die } 3 \leq i)$

Max of *k* dice



- CDF
- ▶ e.g
- CDF
- ▶ pro

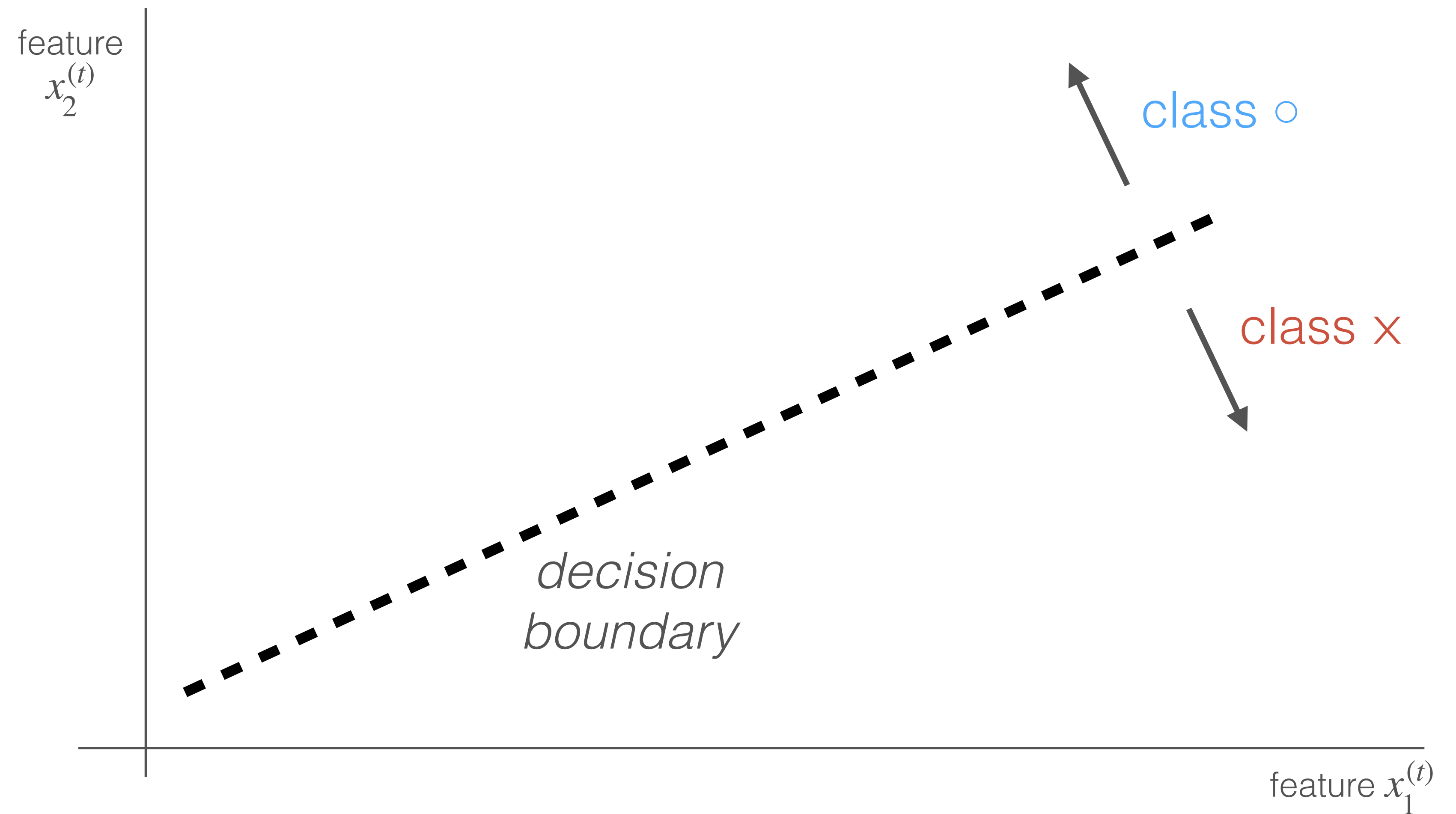
Number of dice	Expected value	Difference
1	3.5000	0.0000
2	4.4722	0.9722
3	4.9583	1.4583
4	5.2446	1.7446
5	5.4309	1.9309
6	5.5603	2.0603
7	5.6541	2.1541

$\leq i)$

What influences the max?

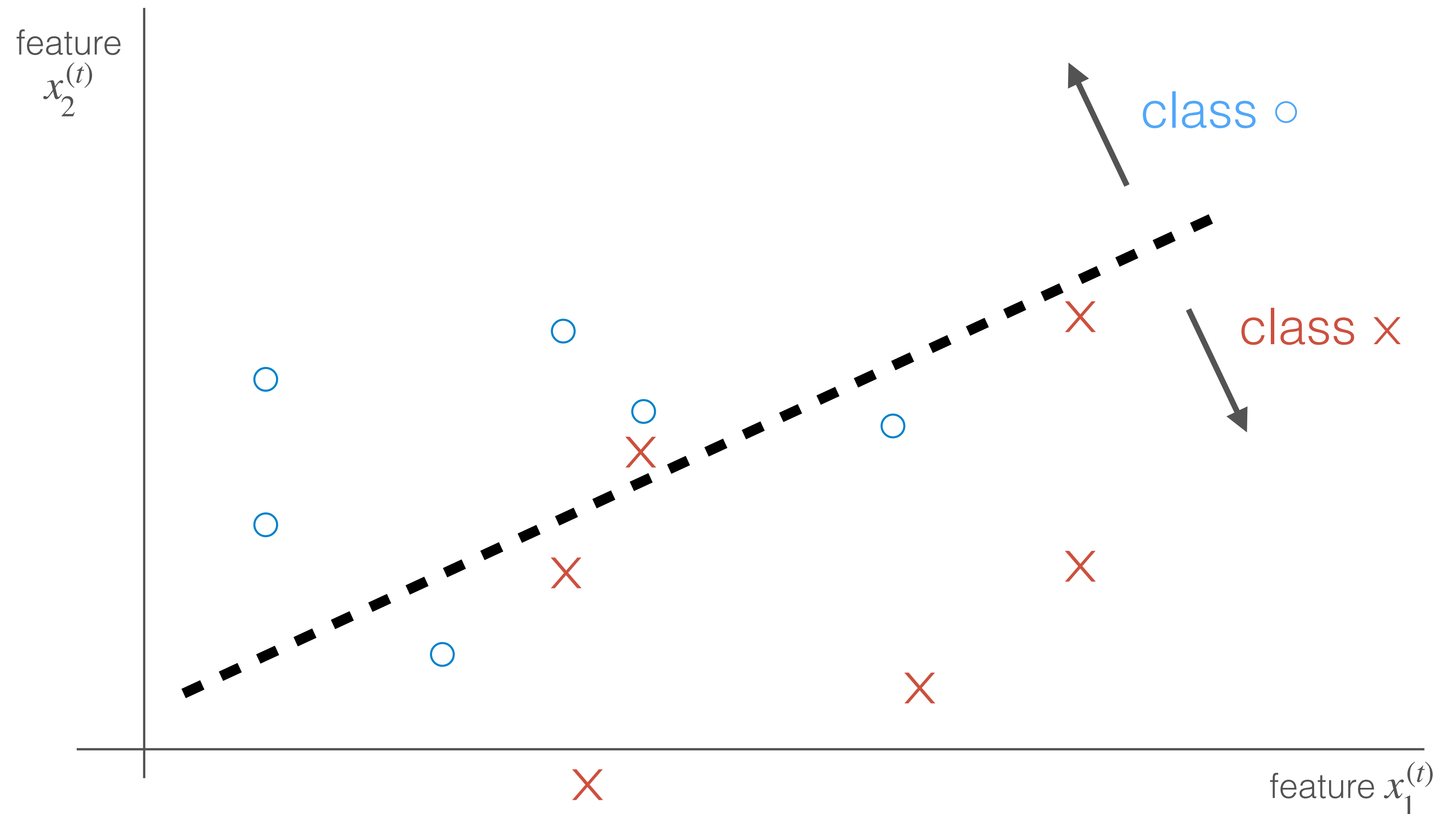
- Just saw: how many dice (more dice = higher)
- What if I add a die that's biased to roll 1 or 2 most of the time?
 - ▶
- What if all the dice are biased so that they roll 1 and 6 much less often?
 - ▶ $\mathbb{E}(1 \text{ die}) =$
 - ▶ $\mathbb{E}(\text{max of } k \text{ dice}) =$

An experiment



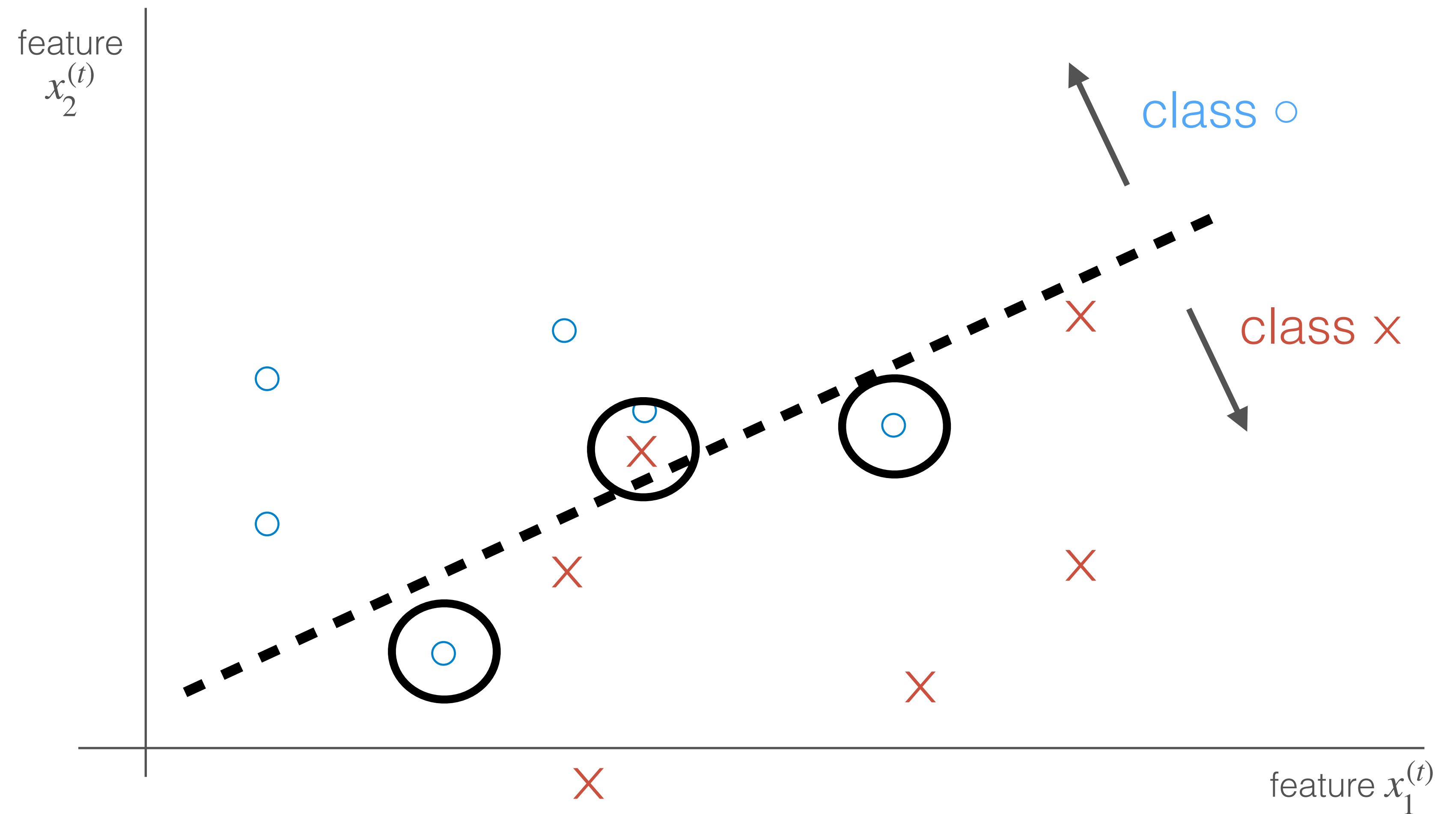
- Fix a model (here: a linear classifier)

An experiment



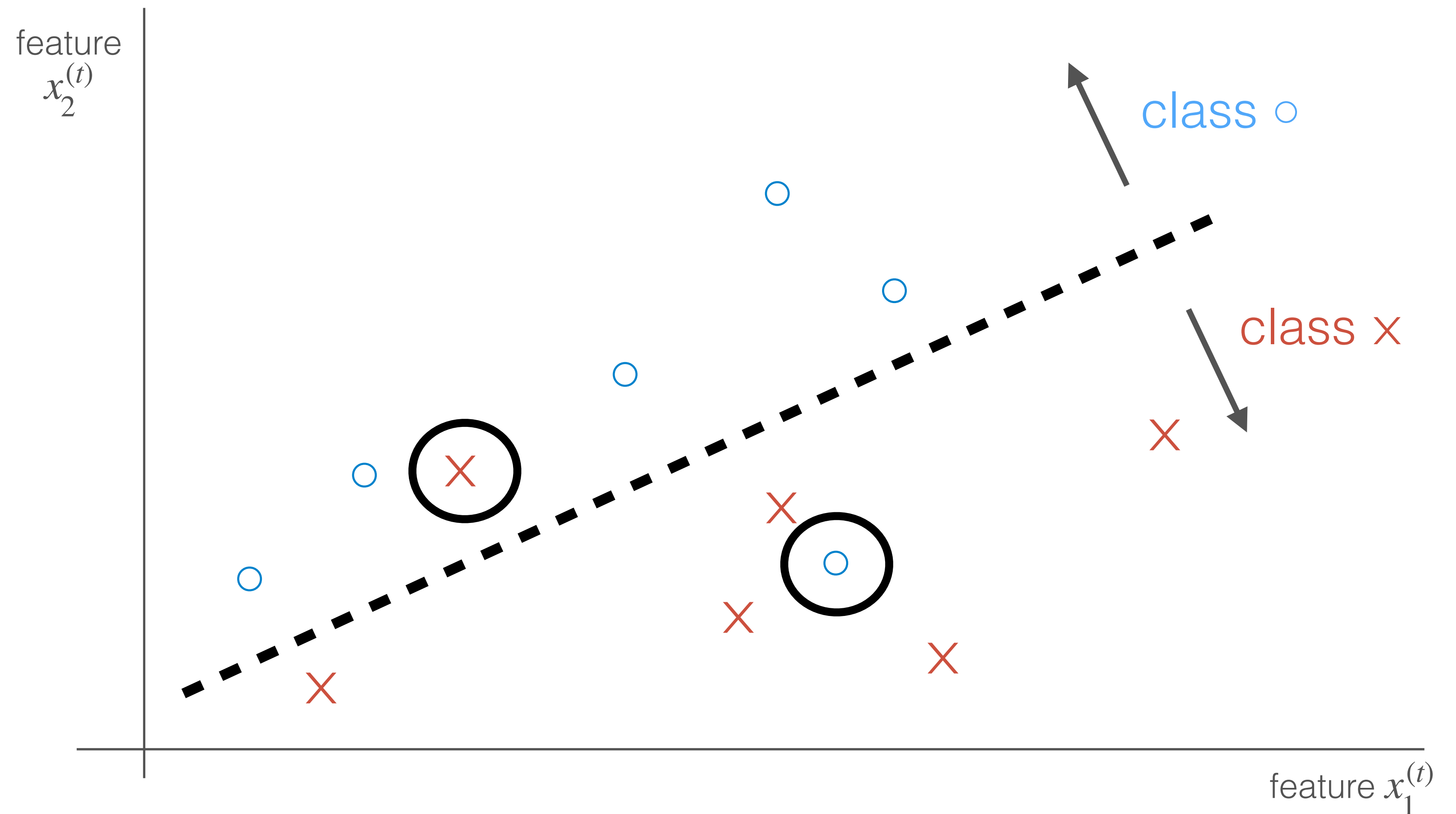
- Sample a dataset at random

An experiment



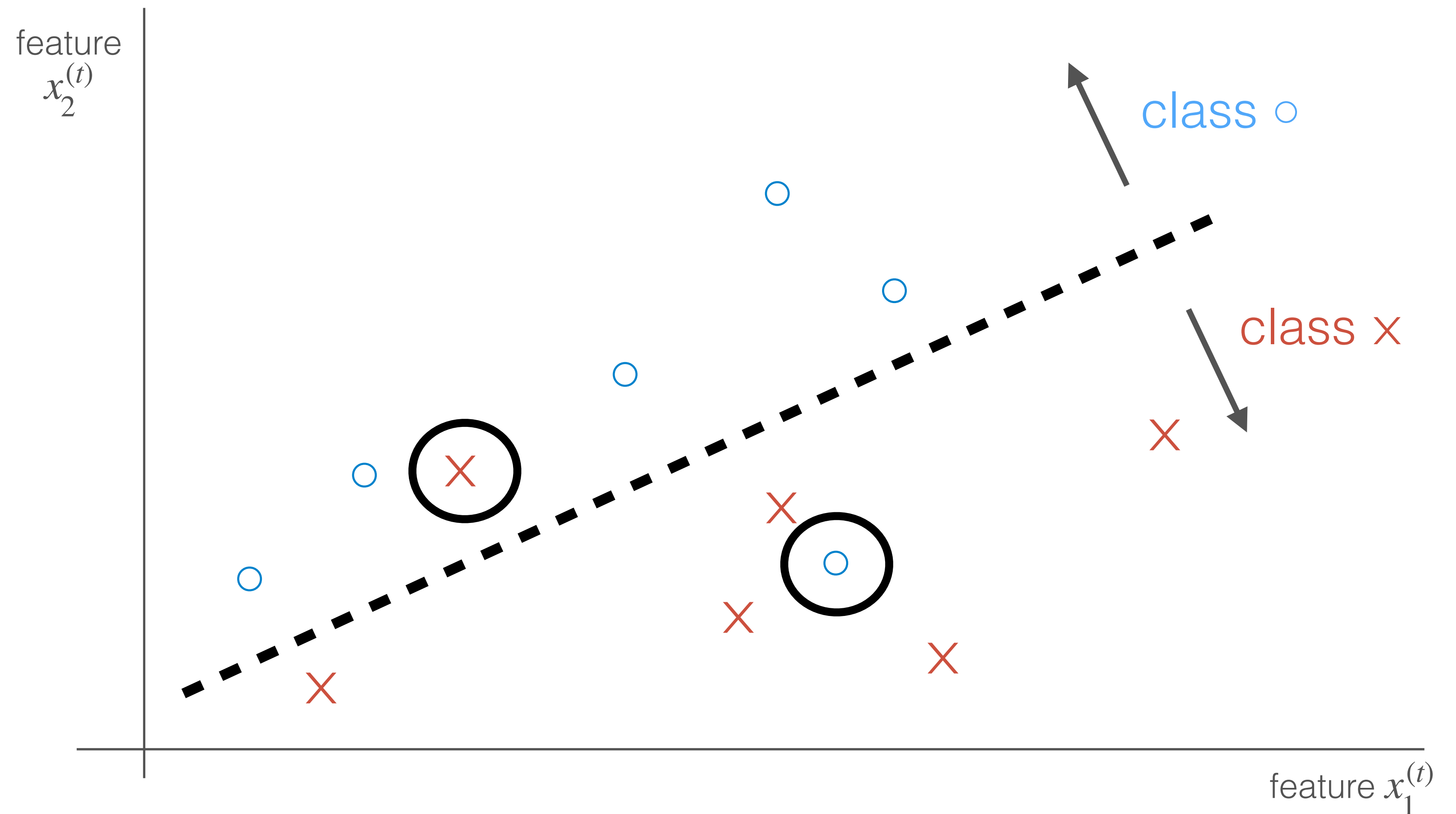
- Measure error rate (here: 3/12, or 25%)

An experiment



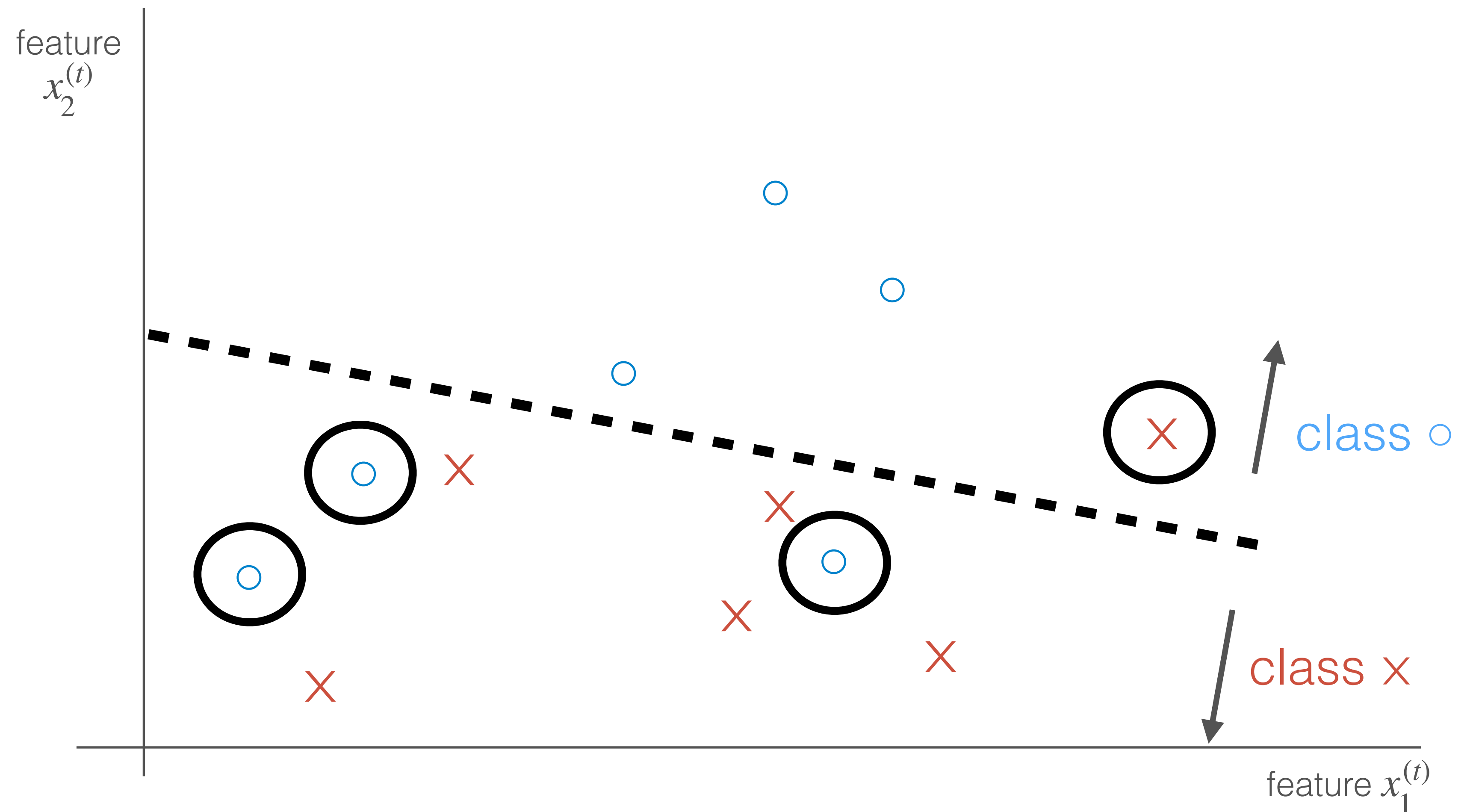
- Result of our experiment is a random variable: different samples can give different error rates (here $2/12 \approx 17\%$)

An experiment



- $\mathbb{E}(\text{error})$ is called the **true error rate** of the classifier
- Variance of measured error depends on the size of the training set

Another experiment



- Evaluate a different classifier on the same dataset: 33%
- Now have two random variables: each time we sample a dataset we can measure error for both classifiers

Now: train a model

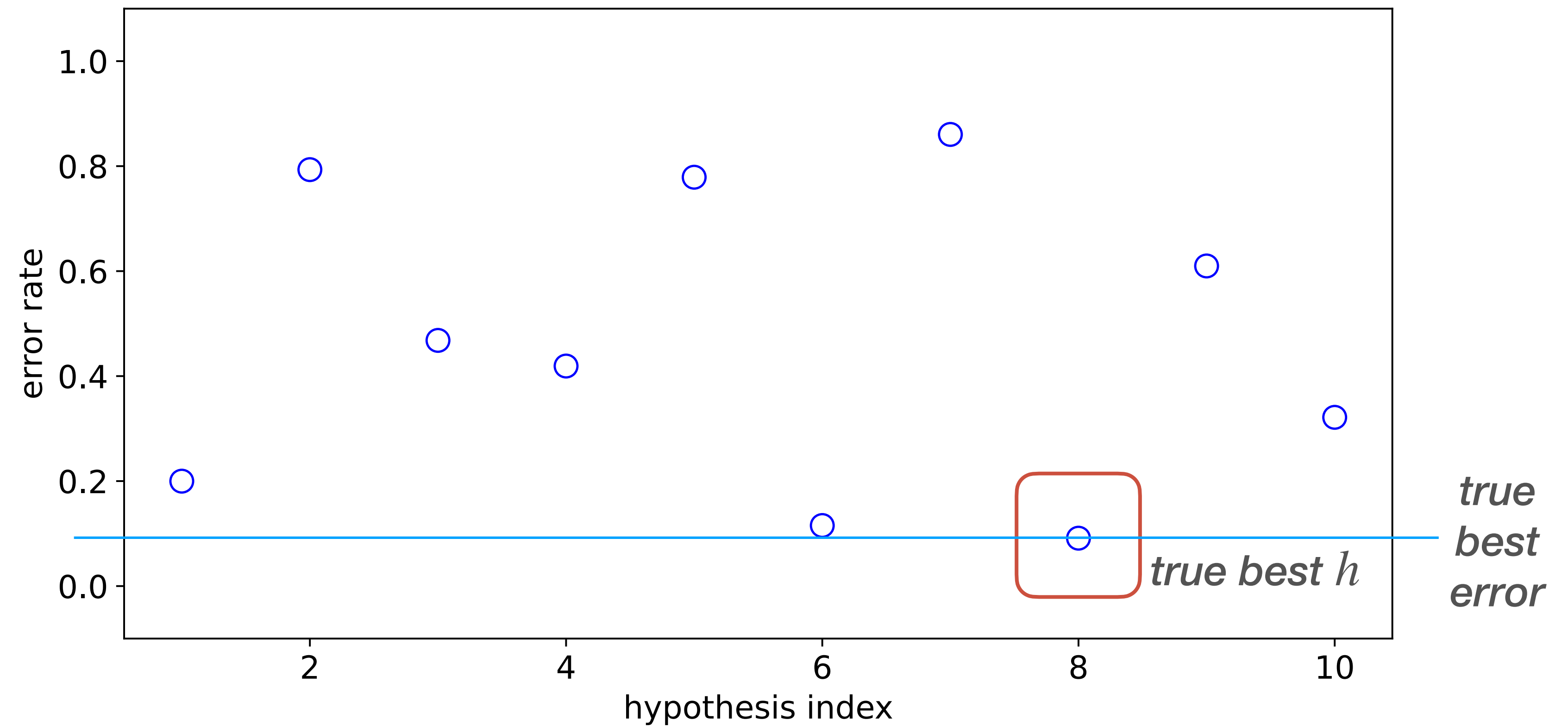
- Sample a dataset at random — call it our training set
 - Evaluate many classifiers (every $h \in \mathcal{H}$) on this dataset
 - Choose the one that looks best (lowest measured error)
-
- This should remind you of our previous experiment



Analogy

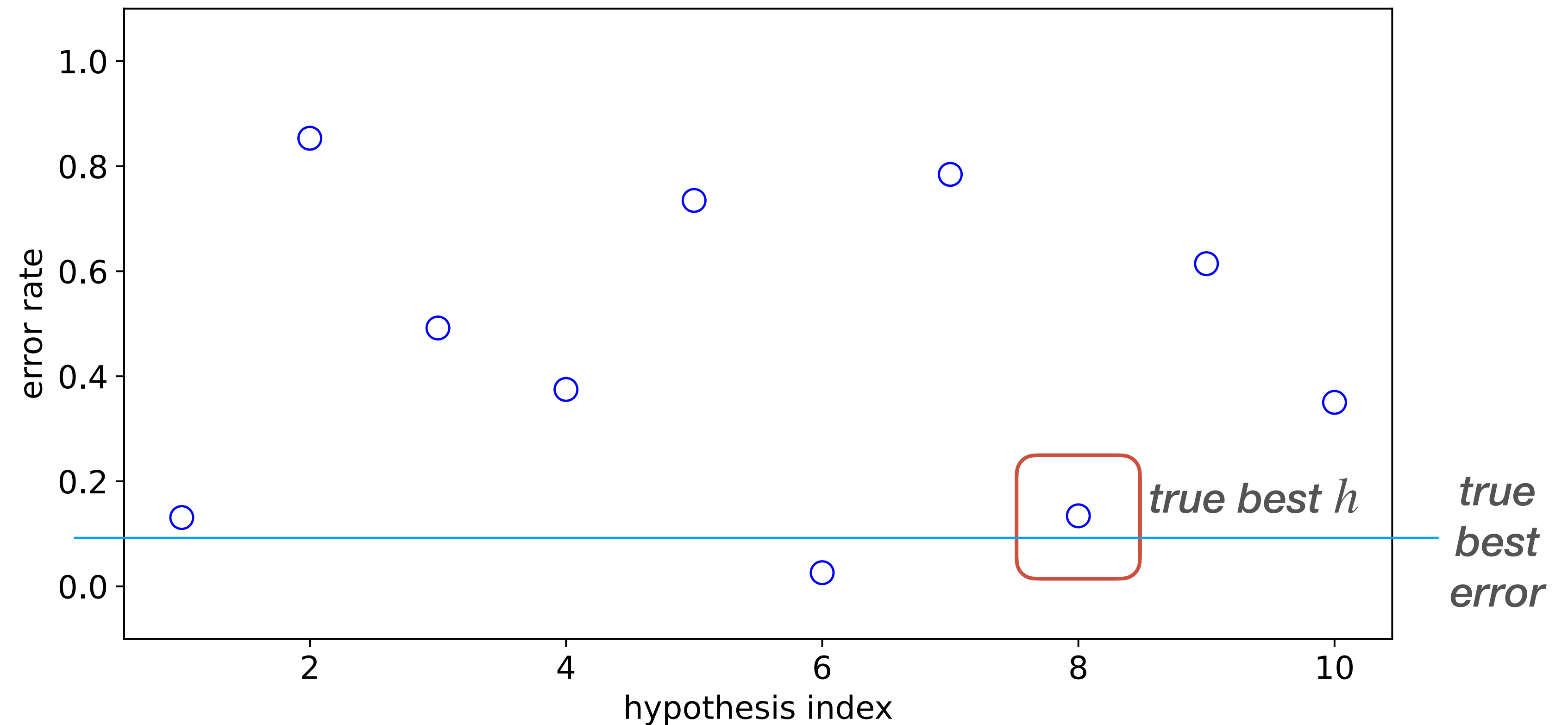
- Evaluating a single model on a random dataset is like rolling a die — a random number, depending on how lucky the model is (how often we select training points that the model happens to do well on)
- Evaluating many models is like rolling many dice: many random numbers, with some models being luckier and some models less lucky
- Training = choosing the apparently-best model from a hypothesis class
 - ▶ like taking the max or min of many dice

Visualize the training process



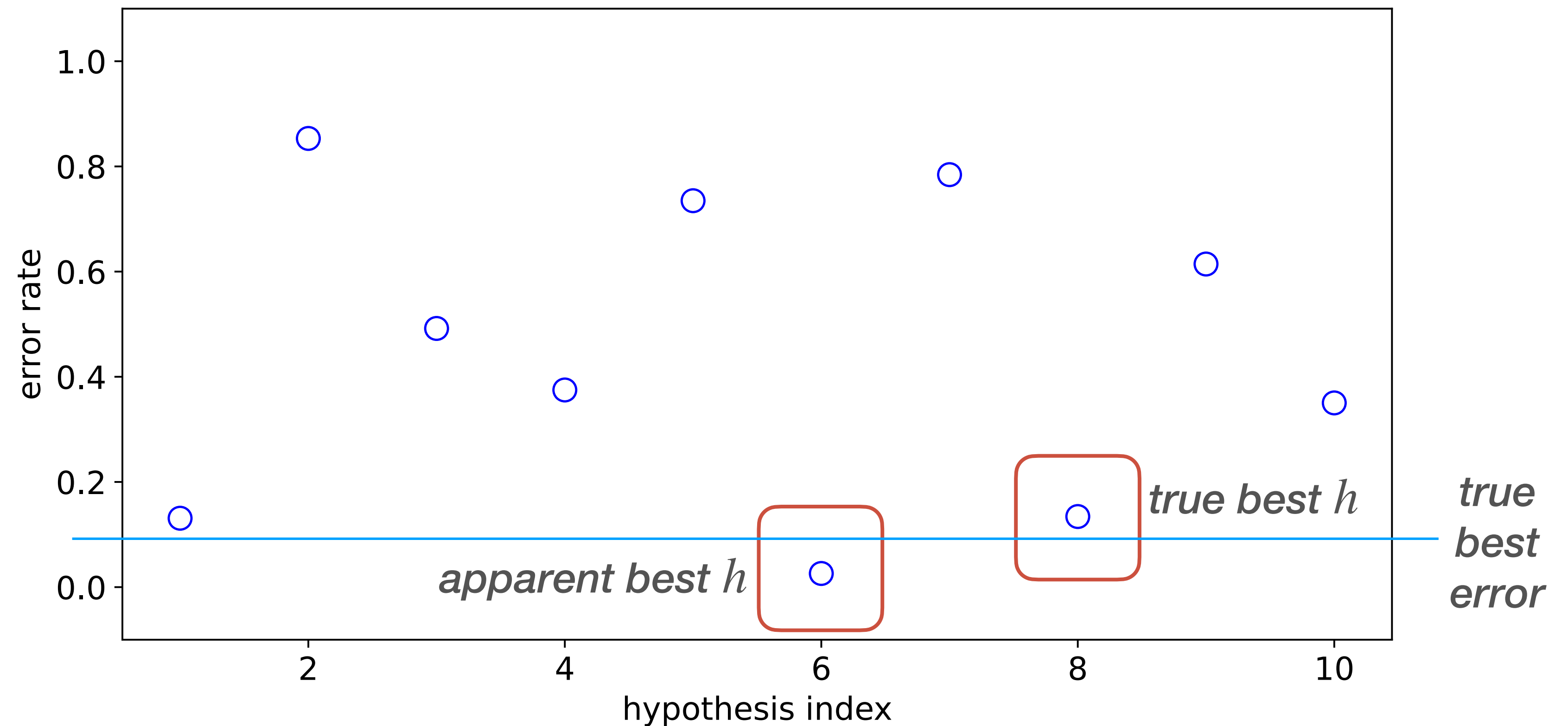
- Suppose we have 10 hypotheses ($|\mathcal{H}| = 10$), each with a different true (unobserved) error

***Draw a
random
training set***



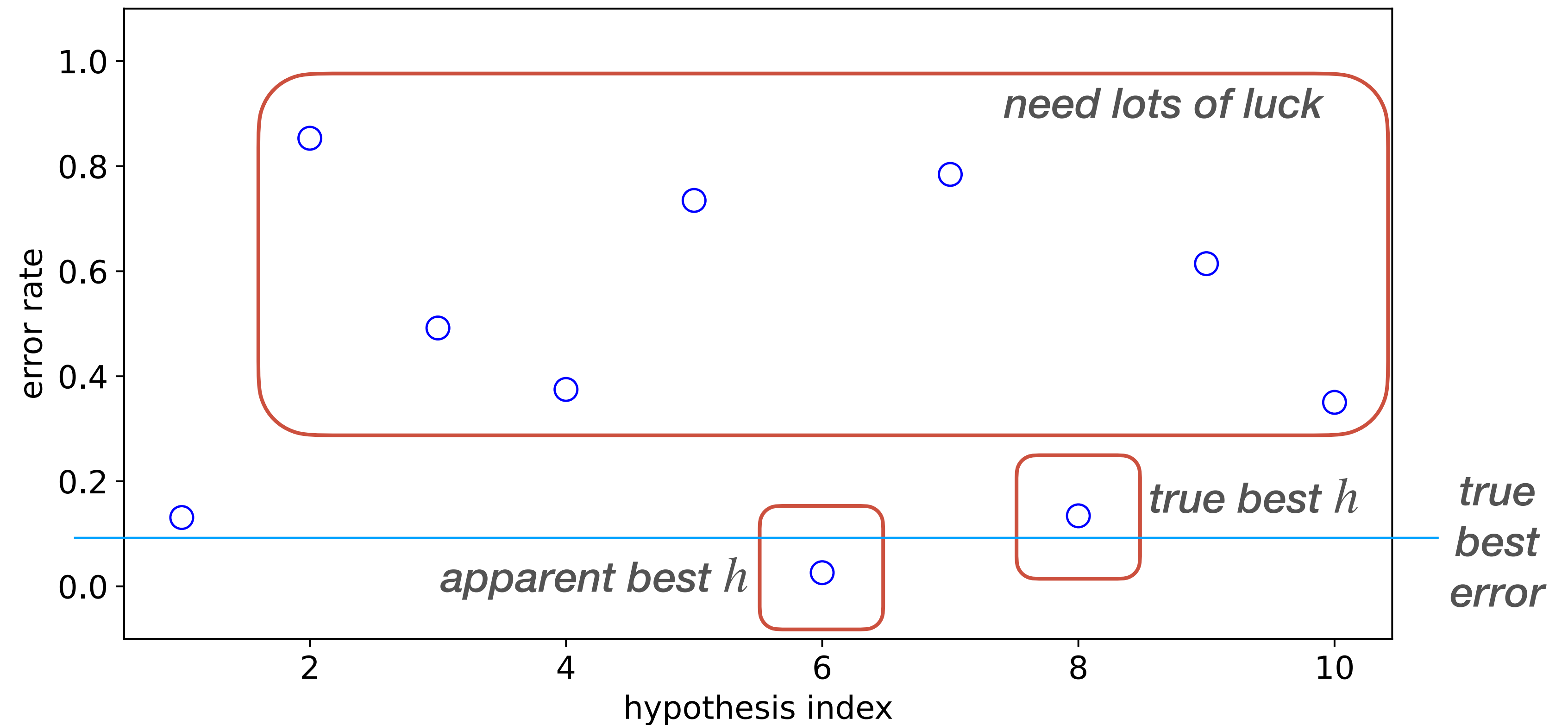
- When we pick a random training set, the measured error for each h is a bit different from its true error — here, best h appears a bit worse than truth

***Pick the
apparently-
best model***



- More importantly, training might get the wrong h !
 - ▶ and its estimated error is better (lower) than the true best error — overfitting!

***Each model
contributes
differently to
overfitting***



- Models with bad true error are individually unlikely to be lucky enough to win — but there can be a lot of them, so even small chances can add up

The problem is max

- Any time we optimize on some examples (train a neural network, select hyperparameters, ...)
- The apparent accuracy of the winner on those examples will be higher than its true accuracy
 - ▶ “winner’s curse,” “selection bias,” “overfitting”
- If we want to know the true accuracy, need fresh examples (a test set)
 - ▶ key is that we didn’t optimize on the fresh examples

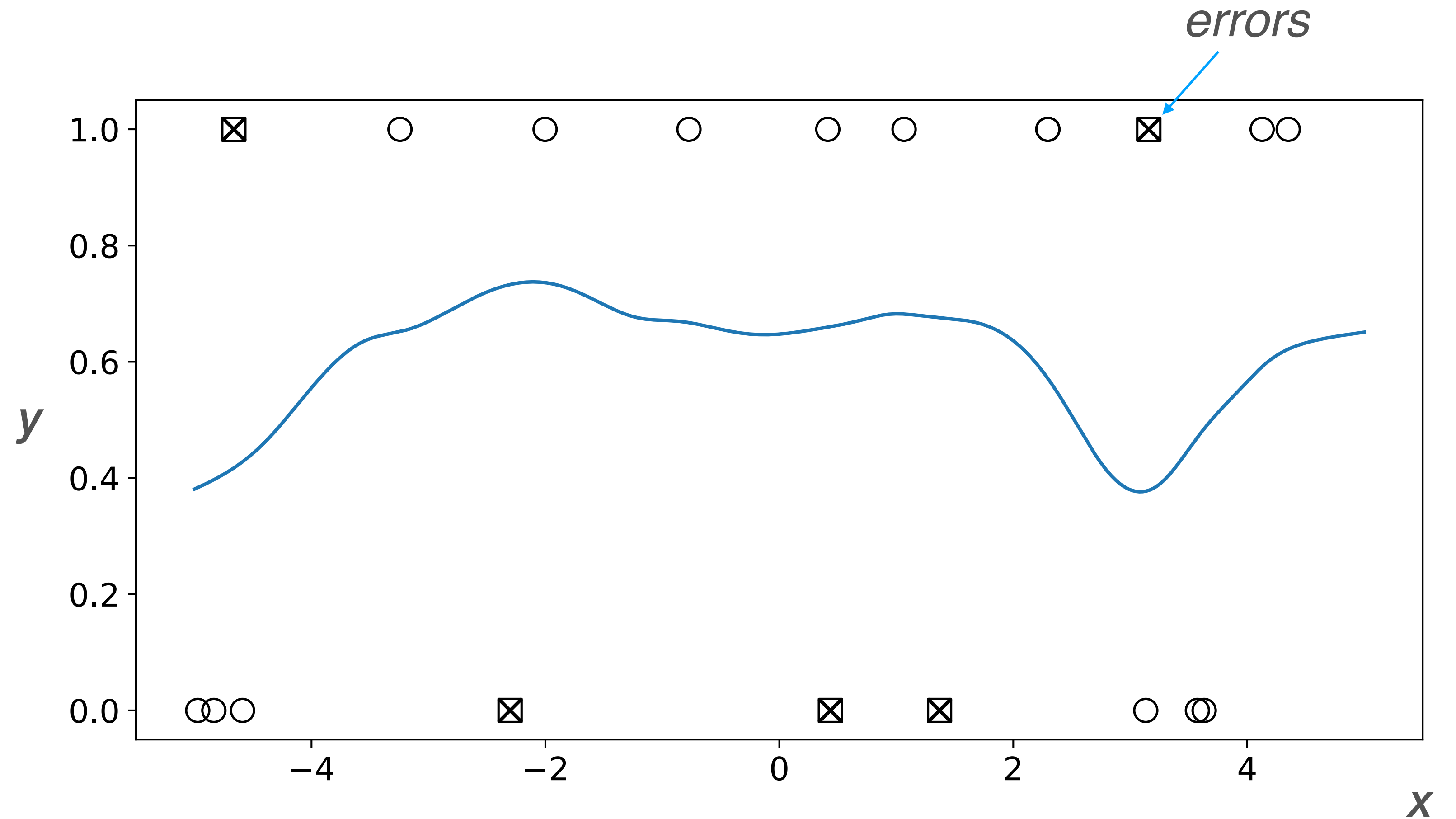
What influences amount of overfitting?

- We can answer by looking at our dice analogy
 - We gave three factors that influence dice experiment:
 - ▶ number of dice
 - ▶ how many of them are close to the true optimum
 - ▶ how big is the variance of each die roll
 - Similarly, for training a model
 - ▶ how many distinct models / complexity of $\mathcal{H}$
 - ▶ how many distinct models are near optimal
 - ▶ how much error in each model evaluation / how big is our training set, how much variance in each training label
- distinct models =
(near-)independent
errors*

Overfitting example

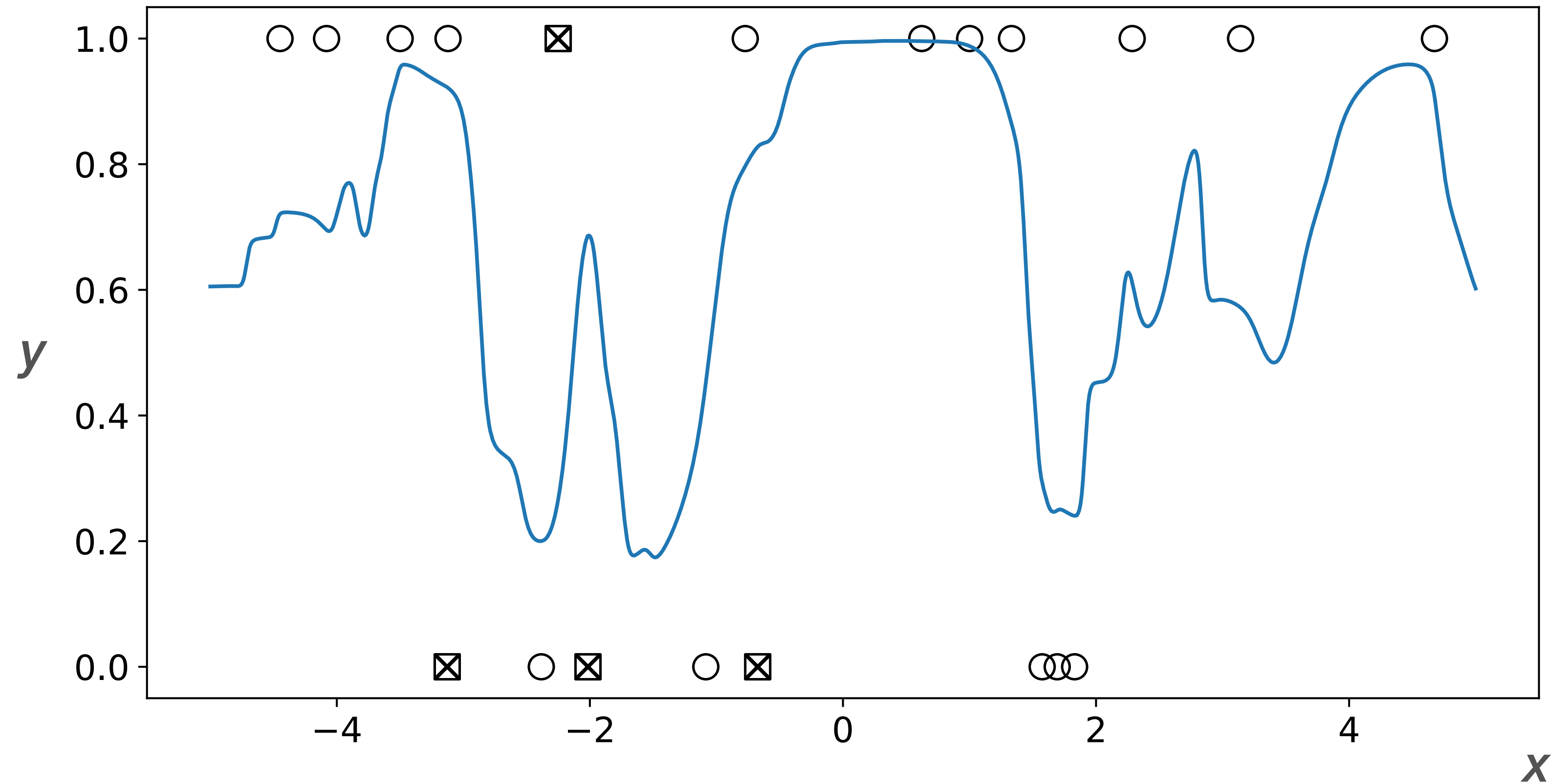
- Fix a training set: 1d classification, $N = 20$ examples, balanced classes (half are class 1, half are class 0)
 - ▶ true (unknown) $h^*: p(y^{(t)} = 1) = \frac{1}{2}$, independent of $x^{(t)}$
- Fix a hypothesis class: $H = 5000$ randomly-initialized neural networks: depth 3, width 32, tanh activation
- Train: choose the network in $\mathcal{H}$ with lowest training error (threshold output $p(y^{(t)} \mid x^{(t)}, w)$ at 0.5)
 - ▶ note: if we trained the usual way (optimized weights w/ PyTorch, without restricting to a finite set of networks) we would get a perfect fit (0% training error)
 - ▶ we are artificially limiting complexity of $\mathcal{H}$, both to limit overfitting and to better match our analysis above

Result



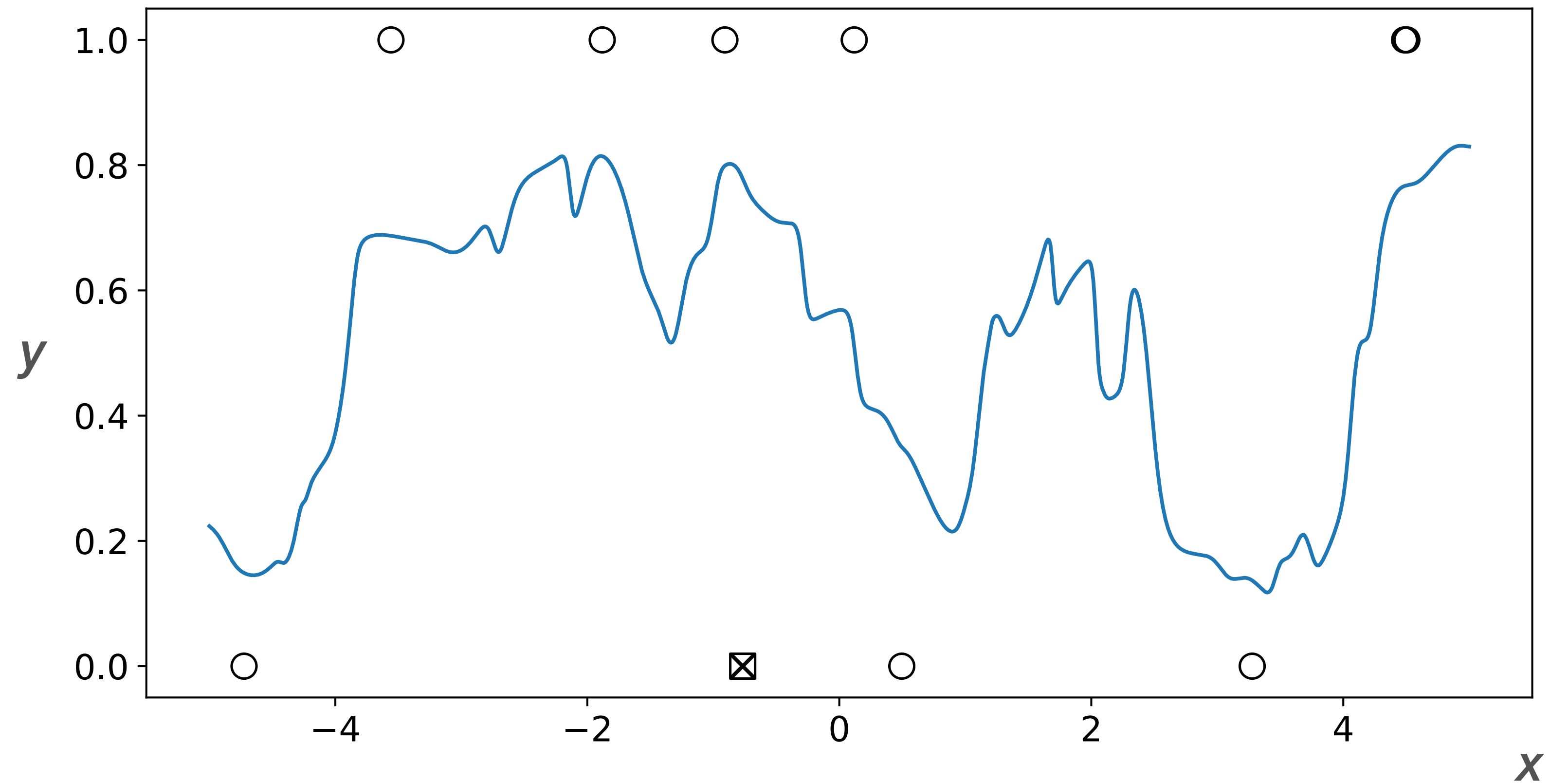
- Training error: $5/20 = 25\%$
 - ▶ cf. true error 50%: overfitting by 25 percentage points

Result



- If we change init parameters to decrease correlation between adjacent predictions, we get more overfitting (here $4/20 = 20\%$ training error)

Result



- If we reduce size of training set (here $N = 10$) for same H , we get even more overfitting (error $1/20 = 5\%$)

How to limit overfitting?

- Get more training data
- Limit model complexity (effective count of models in $\mathcal{H}$)
 - ▶ limit $|\mathcal{H}|$ for a finite model class
 - ▶ limit number of nonzero weights in a linear model
 - ▶ limit precision of weights (find a group of similar models that all do OK)
 - ▶ limit bit-length of model description
 - ▶ regularize: limit norm of weight vector
- These don't always work for modern models!
 - ▶ for example, often see that increasing number of weights in a deep net will *decrease* overfitting
 - ▶ need new measures of simplicity: large margin, low sharpness, few training iterations — active research area

Learning objectives — overfitting

- What is overfitting? What is underfitting? How can we recognize them?
- What are training error and true error, and how do they differ?
- Why does overfitting happen?
- How does $\mathbb{E}(\max\{\text{some r.v.s}\})$ compare to $\max(\mathbb{E}(\text{some r.v.s}))$?
- What factors influence the amount of overfitting?
- What can we do to limit overfitting?

Information



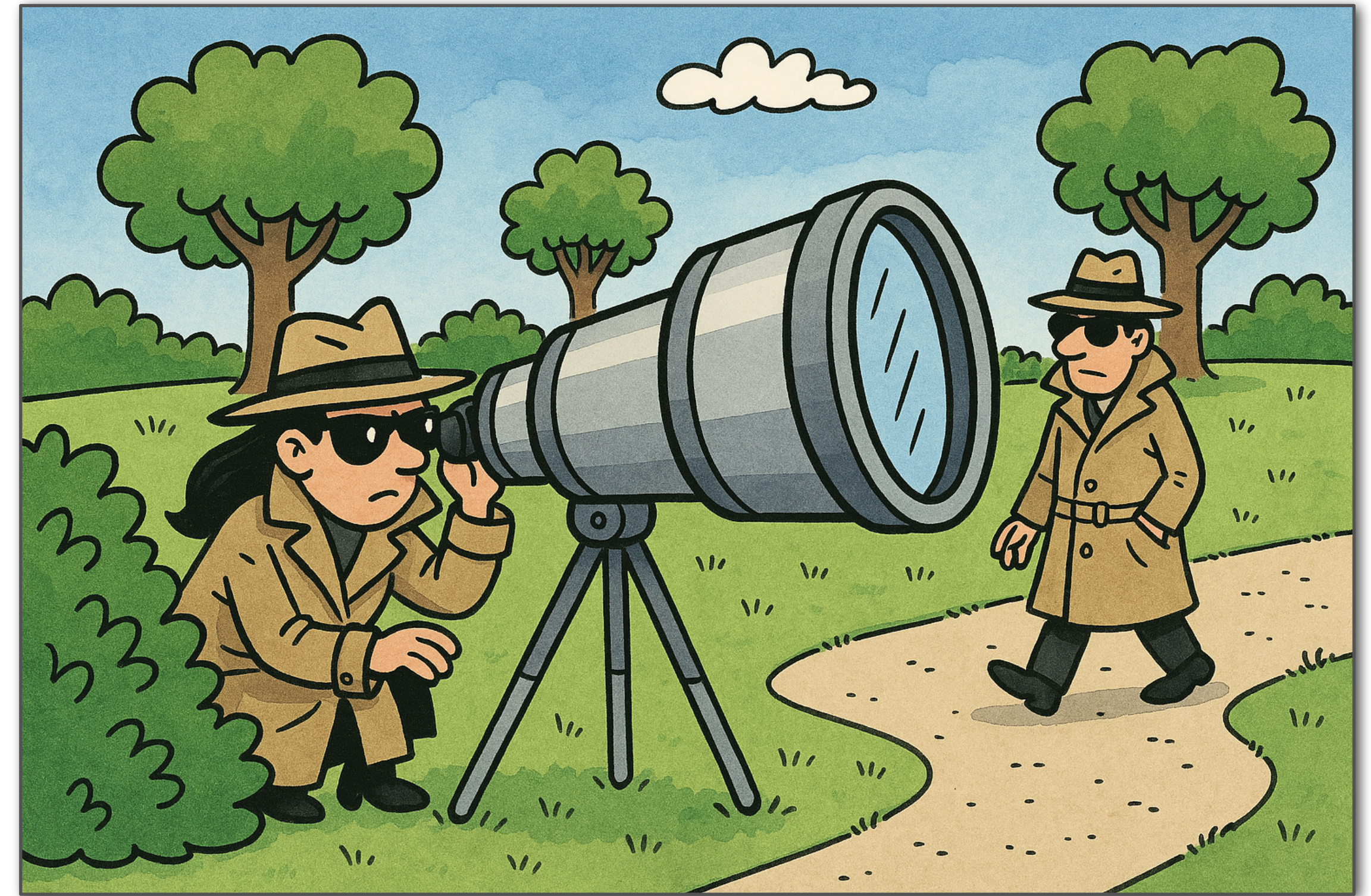
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Complexity $\approx$ ***information***

- We want to limit complexity of our learner
- One general way to measure complexity is ***information***
 - ▶ a simple learner is one that only needs a small number of bits to specify
 - ▶ e.g., a linear classifier with only a few nonzero weights, and where each weight needs only low precision
- Of course we don't just want the learner to be simple — also want it to predict stuff
- So, can think of learning as ***compression***: communicate lots of bits about our data (accurate labels for many examples) using only a few bits of hypothesis

Communication channel

- Every morning, Bob finds out a new secret fact: he discovers what the prime minister had for breakfast
- Immediately, he walks in the park, sits at a specific bench, and arranges the rocks next to it in one of four patterns
- Alice observes through a telescope and learns Bob's message. She informs her superiors.



Code book

meaning	code word	
pancakes	00	
shredded wheat	10	
bagels	01	
caviar	11	

- This channel has a *rate* of 2 bits per day
 - ▶ we could speed it up by having more arrangements of rocks, having Bob take multiple walks per day, etc.
- Whatever the rate is, we want to use it efficiently: send as much information as possible per bit
 - ▶ for this purpose, A&B set up a *code book* ahead of time, like the one above for PM's breakfast items

Uniform code book

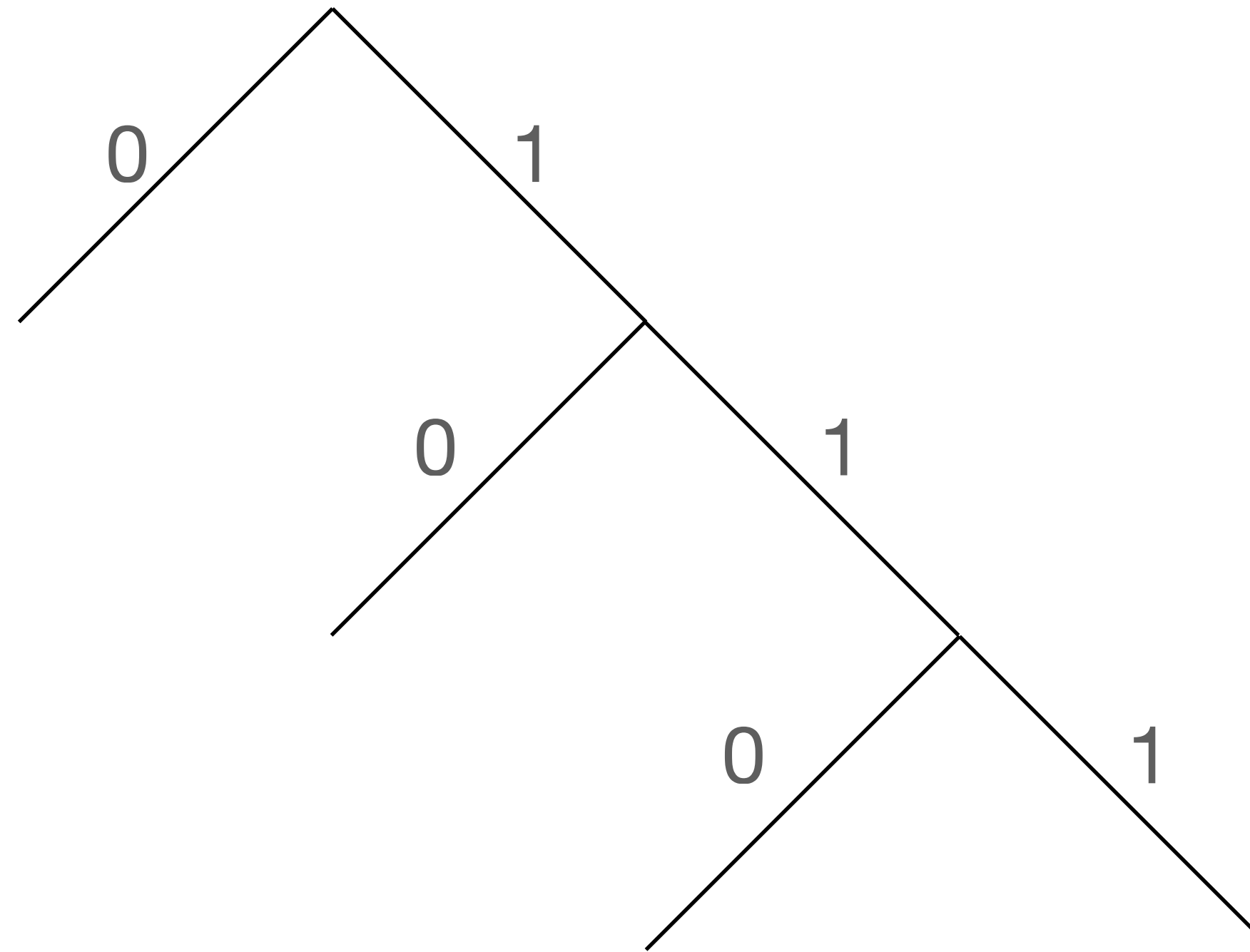
- In this book, all code words are the same length: if there are N possible breakfasts, each is $\lceil \log_2 N \rceil$ bits
- This is optimal in the worst case, which turns out to be when A&B expect all messages to occur equally often (uniform distribution)
- If some events are more common, we might want to send shorter messages for them

Nonuniform code book

meaning	code word	probability
pancakes	0	$\frac{1}{2}$
shredded wheat	10	$\frac{1}{4}$
bagels	110	$\frac{1}{8}$
caviar	111	$\frac{1}{8}$

- An example nonuniform code book (note *prefix property*)

Prefix property



Can guarantee no codeword is a prefix of another by making them paths to leaves in a binary tree

If prefix property failed (e.g., if 11 were a word), we'd be stuck after seeing 11: is this the end or do we have to wait for one more bit?

Nonuniform code book

meaning	code word	probability
pancakes	0	$\frac{1}{2}$
shredded wheat	10	$\frac{1}{4}$
bagels	110	$\frac{1}{8}$
caviar	111	$\frac{1}{8}$

- An example nonuniform code book (note *prefix property*)
- Expected message length = $p_1b_1 + p_2b_2 + \dots + p_Nb_N$:
here $1\frac{3}{4}$
 - ▶ compression! this is what a good codebook buys us
- But if we were wrong and outcomes really were uniform,
expected length would be $(1 + 2 + 3 + 3)/4 = 2\frac{1}{4}$

e.g., if $p_i = \frac{1}{4}$, need 2 bits

Shannon bound

- Lower bound (Shannon): best code uses $\geq \log_2 \frac{1}{p_i}$ bits on average to send a message whose probability is p_i
 - ▶ in fact we can achieve exactly this value in the long run, but have to be slightly cleverer than a fixed code book
 - ▶ or hope that all p_i are integer powers of 2
- Example of cleverer method: arithmetic coding

Entropy

- Sum over all possible messages: expected message length is $-\sum_i p_i \log_2 p_i$, the entropy of distribution p
 - ▶ this is binary entropy; if we send digits instead of bits we get base-10 entropy (using $\log_{10}$)
 - ▶ also common to report *nats* (using $\ln$)
 - ▶ differ by constant factor: more information per symbol $\leftrightarrow$ fewer symbols sent
 - ▶ e.g., digits vs. bits: factor $\log_2 10 \approx 3.32$ fewer symbols (just like writing a number in decimal vs. binary needs fewer symbols: 345 vs. 101011001, 3 digits vs. 9 bits)
- Entropy = how many bits to communicate one sample of a random variable

Wrong codebook

- Using a codebook based on q when true distribution is p costs $-\sum_i p_i \log_2 q_i$
 - ▶ messages are $-\log_2 q_i$ bits instead of $-\log_2 p_i$ bits
 - ▶ expectation over true distribution p
 - ▶ e.g., A&B set codebook before knowing true $p(\text{breakfast})$, can't meet up to fix it
- Called the ***cross entropy*** between p and q
 - ▶ cost of a codebook based on wrong distribution

Kullback-Leibler divergence

- Difference: an extra $\sum_i p_i \log_2 \frac{p_i}{q_i}$ bits
 - ▶ always ≥ 0 , equality iff $q = p$
 - ▶ called $KL(p||q)$, the Kullback-Leibler divergence or relative entropy between p and q
- KL = price (# extra bits) for inaccurate information

Multiple random variables

X	Y	p_{XY}
pancakes	left	$\frac{1}{2}$
shredded wheat	left	$\frac{1}{6}$
pancakes	right	$\frac{1}{6}$
shredded wheat	right	$\frac{1}{6}$

X = breakfast,
 Y = side of bed

- Alice and Bob might want to talk about more than one random variable — here breakfast & which side of bed
- Table shows *joint* distribution of X, Y

*Multiple
random
variables —
marginal*

X	Y	p_{XY}
pancakes	left	$\frac{1}{2}$
shredded wheat	left	$\frac{1}{6}$
pancakes	right	$\frac{1}{6}$
shredded wheat	right	$\frac{1}{6}$

X = breakfast,
 Y = side of bed

- We can extract *marginal* distribution of X or Y , like what we were working with before:
 - ▶ e.g., $P(Y = \text{right}) =$ $P(X = \text{pancakes}) =$

*Multiple
random
variables —
conditional*

X	Y	p_{XY}
pancakes	left	$\frac{1}{2}$
shredded wheat	left	$\frac{1}{6}$
pancakes	right	$\frac{1}{6}$
shredded wheat	right	$\frac{1}{6}$

X = breakfast,
 Y = side of bed

- Or, if we find out something (e.g., $Y = \text{left}$), can focus down to **conditional** distribution of remaining variables
 - ▶ e.g., $P(X = \text{pancakes} \mid Y = \text{left}) =$

Information gain

X	Y	p_{XY}
pancakes	left	$\frac{1}{2}$
shredded wheat	left	$\frac{1}{6}$
pancakes	right	$\frac{1}{6}$
shredded wheat	right	$\frac{1}{6}$

X = breakfast,
 Y = side of bed

- Suppose A&B both know the joint distribution above, and B wants to tell A what the PM had for breakfast
 - ▶ in expectation needs $\text{entropy}(q) \approx 0.918$ bits where $q = \left[\frac{2}{3}, \frac{1}{3}\right]$ (the marginal of X)

Information gain

X	Y	p_{XY}	$\left. \vphantom{\begin{matrix} \text{pancakes} \\ \text{shredded wheat} \\ \text{pancakes} \\ \text{shredded wheat} \end{matrix}} \right\} p = \left[\frac{3}{4}, \frac{1}{4} \right]$
pancakes	left	$\frac{1}{2}$	
shredded wheat	left	$\frac{1}{6}$	
pancakes	right	$\frac{1}{6}$	
shredded wheat	right	$\frac{1}{6}$	

- What if B knows that PM got out of bed on left?
 - ▶ true distribution of breakfast is now p = conditional of X given left — but A doesn't know this, so B still has to use codebook for marginal q
 - ▶ pay an extra $D_{KL}(p||q) \approx 0.024$ bits on average
 - ▶ alternate way to say it: if Bob can tell Alice that PM got out of bed on left, she will gain 0.024 bits of information about breakfast (need 0.024-bit shorter message)

Information gain

X	Y	p_{XY}	} $p' = \left[\frac{1}{2}, \frac{1}{2} \right]$
pancakes	left	$\frac{1}{2}$	
shredded wheat	left	$\frac{1}{6}$	
pancakes	right	$\frac{1}{6}$	
shredded wheat	right	$\frac{1}{6}$	

- What if B knows that PM got out of bed on right?
 - ▶ true distribution is now p' ; pay an extra $D_{KL}(p' || q) \approx 0.085$ bits from having to use q 's codebook
 - ▶ alternate way to say it: if Bob tells Alice that PM got out of bed on right, she gains 0.085 bits of information

Mutual information

X	Y	p_{XY}
pancakes	left	$\frac{1}{2}$
shredded wheat	left	$\frac{1}{6}$
pancakes	right	$\frac{1}{6}$
shredded wheat	right	$\frac{1}{6}$

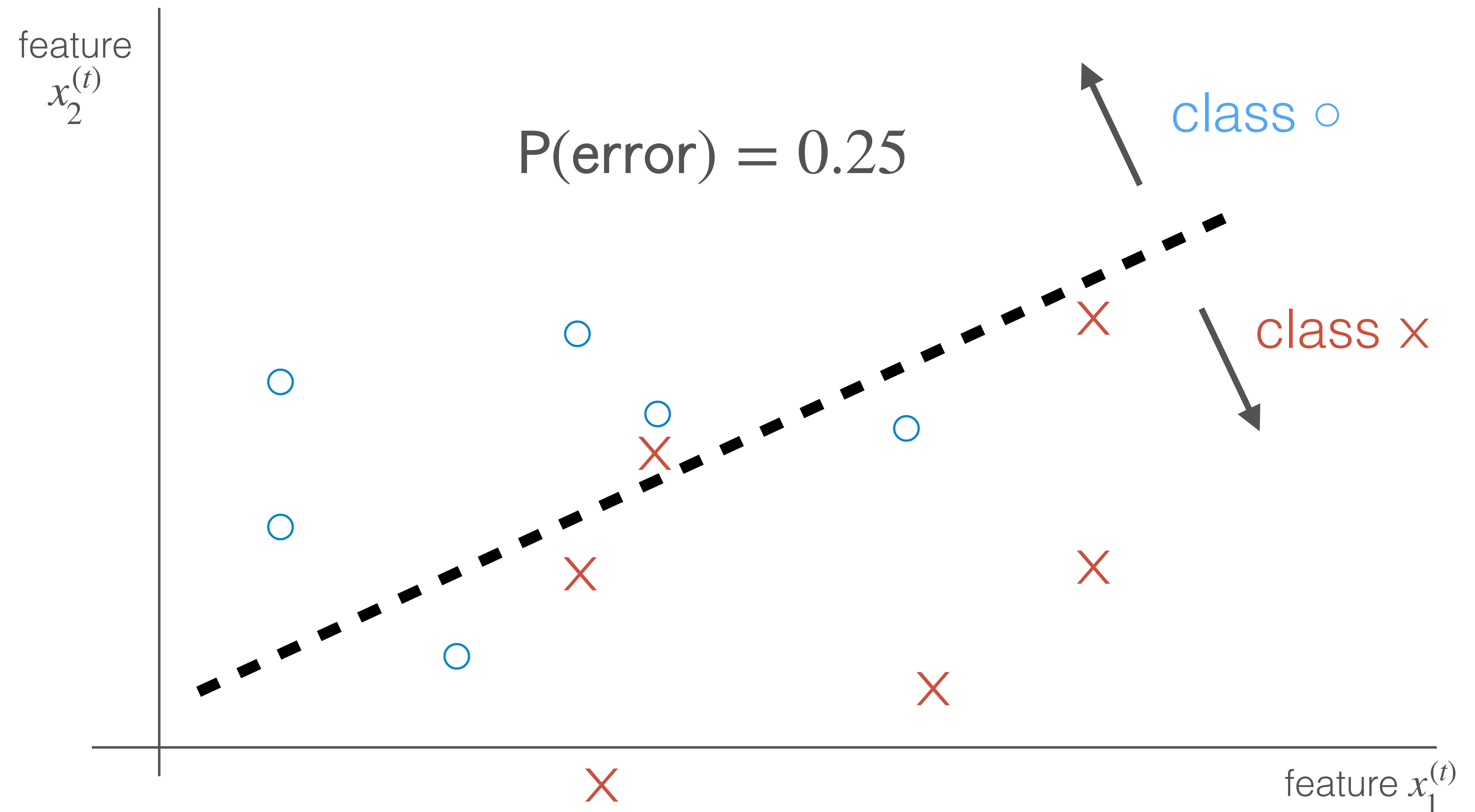
- On average, if B tells A left or right each day, A gains 0.024 bits on $\frac{2}{3}$ of days, and 0.085 bits on $\frac{1}{3}$ of days
- These 0.044 bits are the expected information gain from finding out side-of-bed — also called the *mutual information* between side-of-bed and breakfast
- Hard to see from this expression, but MI is symmetric between X and Y

MI = how much does X tell us about Y (or vice versa)

Connect back to learning

- Setup:
 - ▶ both Alice and Bob know $x^{(t)}$ for N examples
 - ▶ but only Bob knows $y^{(t)}$ — wants to send labels to Alice
- Naive strategy:
 - ▶ Bob sends $y^{(t)}$ for all examples, using 1 bit per example
- Possibly better strategy:
 - ▶ Bob learns a model $h \in \mathcal{H}$ (e.g., weight w in $f_w(x^{(t)})$)
 - ▶ first sends h to Alice, then uses h to help send $y^{(t)}$
- A good hypothesis is one that communicates lots of bits about our data using only a few bits of hypothesis
 - ▶ pay once up front, save on each new example

Using hypothesis to help send labels



- Encode the errors instead of the labels
 - ▶ Bob sends training error probability ϵ as part of h
 - ▶ coding cost is $\text{entropy}(\epsilon)$ per label — can be much less than naive cost of 1 bit

Bound

*preview of
upcoming lectures
on learning theory*

- Can even make a bound: whatever compression we see for our training set, we're likely not to do much worse on unseen future test examples
- Whp, coding cost per example on training set satisfies

$$\begin{array}{ccccc} \text{cost under true} & & & & \text{terms that} \\ \text{error rate} & \leq & \text{observed cost} & + & \text{shrink w/ N} \end{array}$$

- Observed cost per example uses observed training error rate, and includes (amortized) cost of sending h
- LHS grows w/ true error, so this bound rules out true error being too much larger than observed
- More sophisticated versions of this result actually give nontrivial generalization bounds for modern LLMs — still an active research area

Learning as compression

- Weird but fairly decent learning algorithm: `gzip`
- Training:
 - ▶ collect positive-class examples into `pos.txt`
 - ▶ collect negative-class examples into `neg.txt`
- Prediction for a new example `test_ex.txt`:
 - ▶ get file size of `cat pos.txt test_ex.txt | gzip`
minus file size of `gzip pos.txt`
 - ▶ similarly, size of `cat neg.txt test_ex.txt | gzip`
minus size of `gzip neg.txt`
 - ▶ pick the class with the ***smaller*** increment in file size
- Learned hypothesis = internal data structures of `gzip`
after processing `pos.txt` and `neg.txt`

Learning objectives— information

- Why is measuring information important to ML?
- What are joint, marginal, and conditional distributions?
- What is the Shannon bound, and what does it tell us about codebooks / message length?
- What happens if we encode a message based on the wrong probability distribution?
- What are entropy, cross entropy, and relative entropy? What is mutual information?
- How does learning relate to communication and compression?